Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** across multiple categories to guide strategic business decisions.

Made with GAMMA

Dataset at a Glance

3,900	18	$59.76	3.75
Transactions	Columns	Avg. Purchase	Avg. Rating

Key Features

- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Category, Amount, Season, Size, Color
- **Behavior:** Discounts, Promo Codes, Frequency, Shipping Type, Review Rating

37 missing values in Review Rating — imputed using category median.

Made with GAMMA

Data Preparation in Python

01

Load & Explore

Imported dataset with `pandas`; used `df.info()` and `.describe()` for initial profiling.

02

Clean & Standardize

Imputed missing Review Ratings; renamed columns to `snake_case`; dropped redundant `promo_code_used`.

03

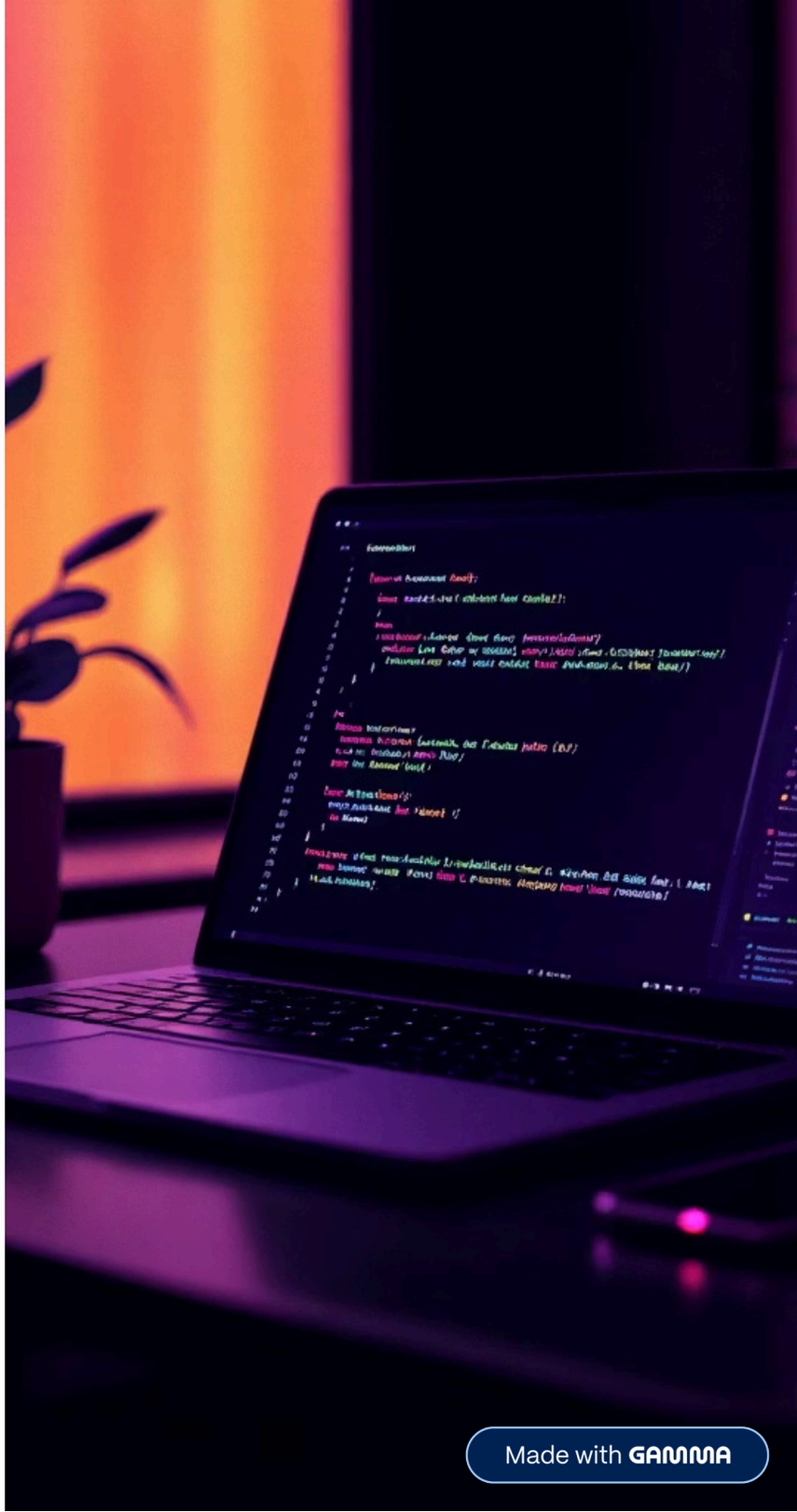
Feature Engineering

Created `age_group` bins and `purchase_frequency_days` from raw purchase data.

04

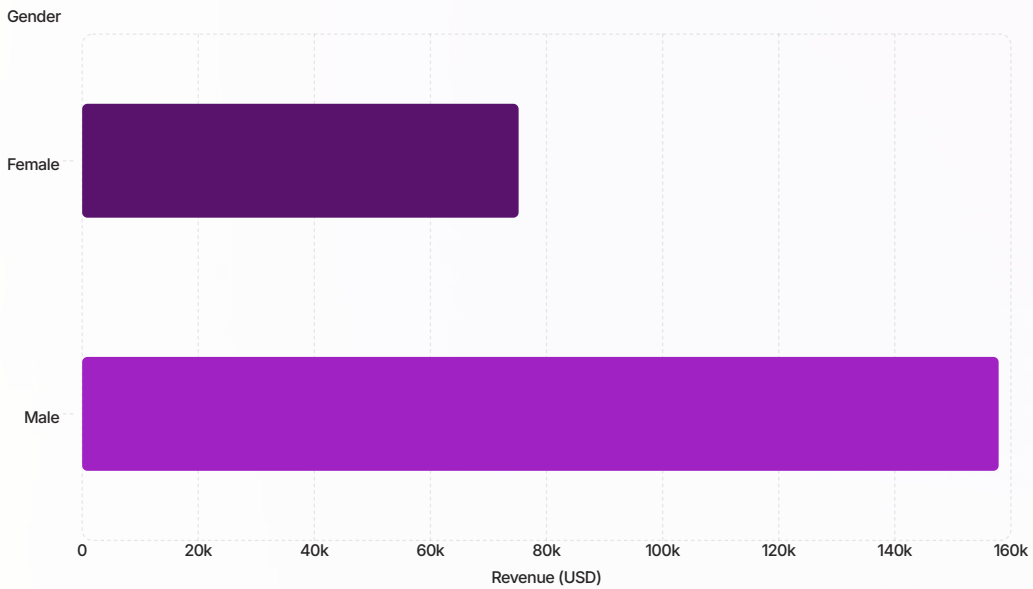
Database Integration

Loaded cleaned DataFrame into **PostgreSQL** for structured SQL analysis.



Made with GAMMA

SQL Analysis: Revenue & Spending



Key Findings

- **Male customers** generated $157,890 vs. $75,191 from female customers.
- **839 discount users** still spent above the average purchase amount — discounts don't always erode value.
- **Express shipping** users averaged $60.48 vs. $58.46 for Standard — a small but notable premium.

Made with GAMMA

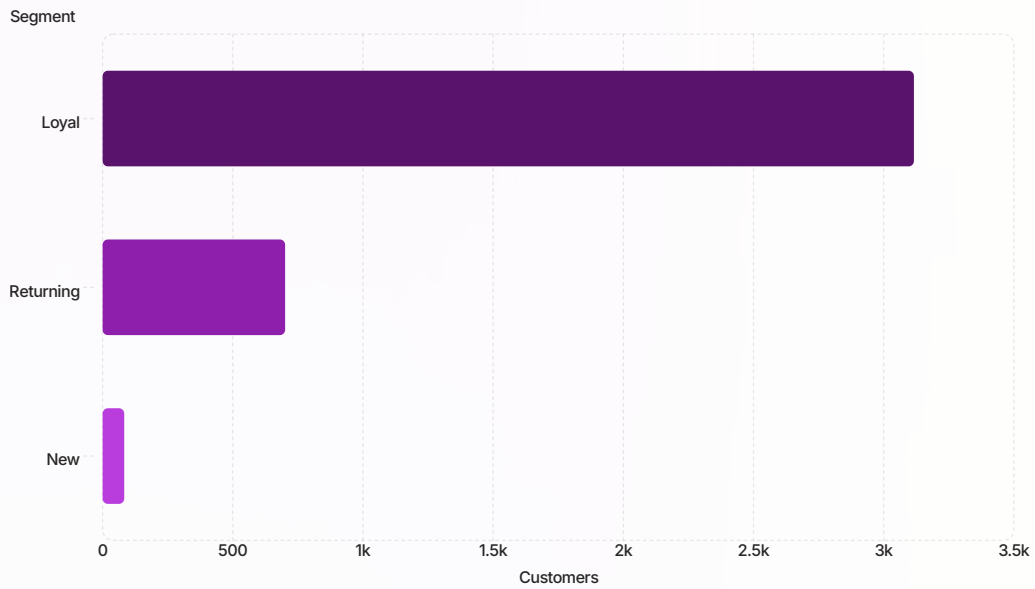
Subscription & Customer Segments

Subscribers (27%)

1,053 customers · Avg spend $59.49 · Total revenue $62,645

Non-Subscribers (73%)

2,847 customers · Avg spend $59.87 · Total revenue $170,436



Made with GAMMA

Product Insights

Top 5 Products by Rating

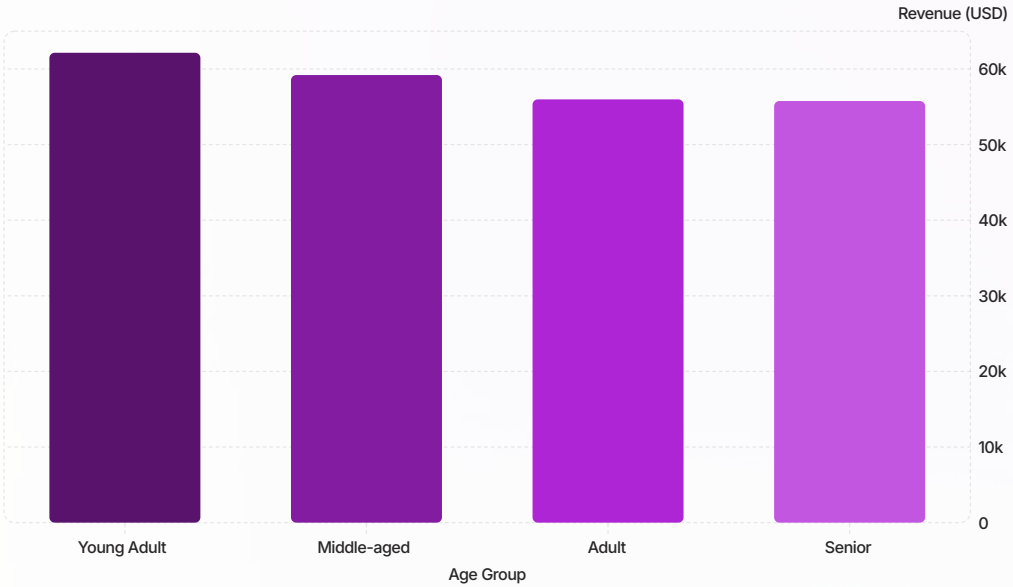
Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

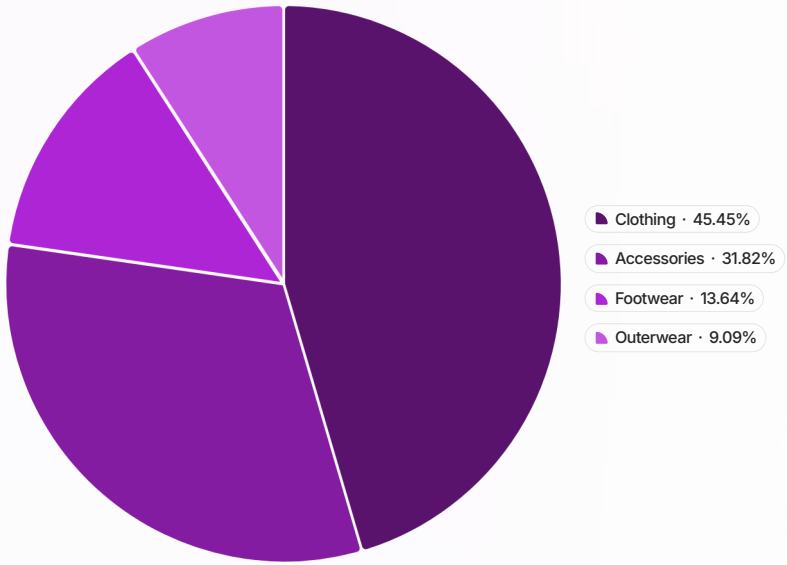
Most Discounted Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Made with GAMMA

Revenue by Age Group & Category

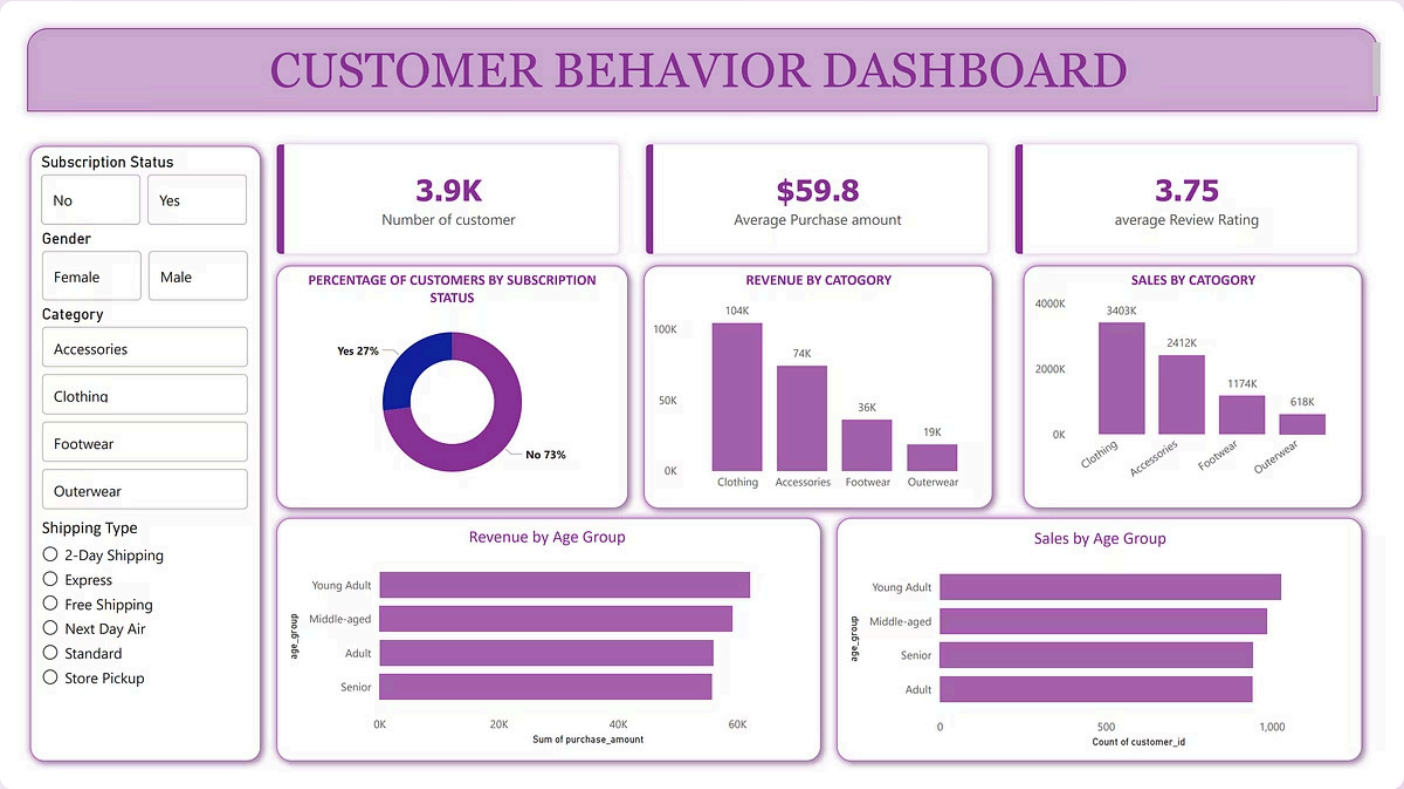




Made with GAMMA

Power BI Dashboard

An interactive **Power BI** dashboard consolidating all key metrics: 3.9K customers, $59.76 avg. purchase, 3.75 avg. rating — with filters by gender, category, and shipping type.



Made with GAMMA

Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscriber base.

Loyalty Programs

Reward repeat buyers to accelerate movement into the "Loyal" segment.

Review Discount Policy

Balance promotional discounts with margin control — especially for high-discount products.

Targeted Marketing

Focus on Young Adults, top-rated products, and express-shipping users for highest ROI.

Made with GAMMA

Key Takeaways

80% Loyal Customers

3,116 of 3,900 customers are in the Loyal segment — a strong retention foundation.

Clothing Dominates

Clothing leads all categories in both revenue (~$100K) and sales volume (~1,800 units).

Subscription Gap

Only 27% are subscribers, yet avg. spend is nearly equal — a major growth opportunity.

Young Adults Lead

Young Adults contribute the highest revenue ($62,143) — the most valuable segment to target.

Made with GAMMA